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77831671648315317851157205990765285234049072855151919714347606167227539176530411895905709620274443857634478715493683675165753254654758209989514774725672337507665072398825386259305
33115897823513490885128049912338627881807786178184506663055910211106407831025359135948691157237088000057972167387473695719543075645935198428517267422697108382457573696201103434081
81451912678845205077154721877485882891435853769575453105694747368449085351078612030300713995402763536887428552533124546180907410715168392932504661103689288996413562990212953597348
7079207957020969779158377956694277675584200801161336006763732002885945463981450198783704094358811192434471002563455248474687473588631728166204025765594273596771266231261269856178

45941757812875343282617739186284923182272009900444797096957065694286487931151376298757382472551880986416384672646579646240450869805000261093409316907075656555897671564003629832899
89184133636425943284208655090735979612173058791448704762390920025290449408218953657635279879172911753821577188852511613120296375507685667611776587064351590012591459112726404047296
55143495006182641972621882619157277902572458928231199048471647051591983912773795694395914612428645843030045298030439417623351024877337547036870090981296865551466521386858825974964
17986717035871120879673404496726769327714963992122180959245853482449697454684208105673484900497325863599120890988049206780085286354074876596598819691078789234261545773239026984196
85532591486739791830984502362079542469757664492825225286809162252480908579147824716696653934312665724699644824764766238935266933924618352687423482846010447982945664535852036474704
41399887095604660338405939614200642282747796750131016860234240203455563710380598949451874640422902385063661334312548842979509359391835461036648721663685423909406525658063889385428
261027081346700499533122074537283113637174283497138815468723361924431890501847692201263248124258358082379681357688196097661118694988663

The Elements

Dominic Alexander Cooper

- 000 I exist
- 001 I feel pain
- 002 My pain can be increased by incompetence
- 003 My pain can be decreased by competence
- 004 I can learn
- 005 Learning increases competence
- 006 Contradiction increases incompetence
- 007 Proof by contradiction does not enable learning
- 008 Proof by error-free construction is learning
- 009 Proof by contradiction is not learning
- 010 Your level of competence is quantifiable
- 011 Your level of incompetence is quantifiable
- 012 Context changes your current level of competence
- 013 The basis span of your competence can be optimized for all contexts
- 014 Everyone is innocent
- 015 Competence is a functional axial n-dimensional space
- 016 Incompetence is not a functional axial n-dimensional space

- 017 Since I exist, I can be treated as a living information-bearing system
- 018 Since I feel pain, pain is a valid system signal
- 019 Since incompetence can increase pain, incompetence is a negative system state
- 020 Since competence can decrease pain, competence is a positive system state
- 021 Since I can learn, my system state can be updated
- 022 Since learning increases competence, learning is a valid update operation
- 023 Since contradiction increases incompetence, contradiction is a corruption signal
- 024 Since proof by error-free construction is learning, construction is the preferred proof engine
- 025 Since proof by contradiction is not learning, contradiction is not a preferred learning engine
- 026 Since competence is quantifiable, competence can be measured
- 027 Since incompetence is quantifiable, incompetence can be measured
- 028 Since context changes competence, competence is context-indexed
- 029 Since competence can be optimized for all contexts, competence has a basis-span
- 030 Since everyone is innocent, correction must not be treated as guilt
- 031 Since competence is functional axial n-dimensional space, competence has dimensions
- 032 Since incompetence is not functional axial n-dimensional space, incompetence lacks stable constructive basis

- 033 The mind can ingest information through perception, language, memory, and action
- 034 The mind can encode information as concepts, habits, images, words, and procedures
- 035 The mind can index information by association, label, context, emotion, and purpose
- 036 The mind can query stored information through attention, recall, questioning, and tool-use
- 037 The mind can update stored information through learning
- 038 The mind can validate stored information through error-free construction
- 039 The mind can reject or quarantine contradictory information when it increases incompetence

```

040 The mind can externalize information into tools, notes, diagrams, calendars, code, and environment
041 The human environment can serve as auxiliary storage
042 Human-made tools can serve as auxiliary memory, index, query, and validation systems
043 The body can store competence as trained procedure
044 Repetition can strengthen retrieval pathways when it increases competence
045 Reflection can detect incompetence when pain, error, or contradiction appears
046 Deliberate practice can transform incompetence into competence
047 A context can be treated as a coordinate region inside competence-space
048 A skill can be treated as an axis inside competence-space
049 A habit can be treated as a repeated function over time
050 A tool can be treated as an external operator that extends mental function
051 A note can be treated as persistent external memory
052 A diagram can be treated as spatial indexing of thought
053 A calendar can be treated as temporal indexing of intention
054 A checklist can be treated as procedural compression
055 A question can be treated as a query operation
056 An answer can be treated as a retrieved or constructed record
057 A mistake can be treated as a detected incompetence marker
058 Correction can be treated as a competence-restoring update
059 Optimization can be treated as basis-span improvement across contexts
060 The mental human-centric database is the union of mind, body, environment, and tools
061 The purpose of the mental human-centric database is to reduce pain by increasing competence
062 The governing rule of the database is to learn by error-free construction
063 The final enforced target is complete competence under {014 + 015}
064 Once true, always, true

```

Below is `MHCDScript v2`, a general-purpose programming language design for the full 64-axiom MHCD system.

The main upgrade from v1 is that v2 treats the MHCD as a state machine with typed records, verified transitions, constructive execution, context-indexed competence, and innocence-preserving correction.

```
# MHCDScript v2
```

```
```mhcd
language MHCDScript version 2
```

```
subject "Dominic Alexander Cooper"
```

```
runtime MHCD {
 principle existence
 principle pain_signal
 principle learning_update
 principle constructive_proof
 principle innocence_preservation
 principle competence_space
 principle context_indexing
 principle tool_extension
}
```
```

```
## 1. Design purpose
```

`MHCDScript v2` is a general-purpose scripting language for directing the human mind, body, environment, and tools as one distributed cognitive database.

Its purpose is to:

```
```mhcd
purpose {
 reduce pain by increasing competence
 detect incompetence without guilt
 reject contradiction as a learning engine
 prefer error_free_construction
 quantify competence and incompetence
 optimize competence_basis across contexts
 enforce {014 + 015}
}
```
```

The governing command is:

```
```mhcd
enforce {
 014: "Everyone is innocent"
 015: "Competence is a functional axial n-dimensional space"
}
```
```

So every program must preserve:

```
```mhcd
error != guilt
mistake == update_requirement
learning == competence_increase
construction == preferred_proof_engine
```
```

2. Full axiom import

Rather than rewriting all 64 axioms inside every script, v2 imports them as a standard library.

```
```mhcd
import MHCD.AXIOMS.000_063
import MHCD.BOOLEAN
import MHCD.LAMBDA
import MHCD.COMPETENCE_SPACE
import MHCD.CONSTRUCTIVE_VALIDATION
```
```

The full axiom set is treated as:

```
```mhcd
axiom_set FULL_MHCD = {
 000..063
}
```
```

Every axiom is internally available as a Boolean truth constant:

```
```mhcd
AXIOM[000] = true
AXIOM[014] = true
AXIOM[015] = true
AXIOM[063] = true
```
```

A program is valid only if it does not violate the enforced axioms:

```
```mhcd
program_valid := respects(014) * respects(015) * rejects(guilt) * prefers(construction)
```
```

3. Primitive types

```
```mhcd
type Bool = 0 | 1
type Number = integer | decimal
type Text = string
type List<T> = ordered collection
type Set<T> = unordered collection
type Map<K,V> = key-value collection

type AxiomID = 000..063
type Subject = living information-bearing system
type Mind = internal meaning processor
type Body = procedural competence store
type Environment = auxiliary cue/storage field
type Tool = external operator
type Note = persistent external memory
type Diagram = spatial index
type Calendar = temporal index
type Checklist = procedural compression
type Query = question operation
type Answer = retrieved or constructed record
type Context = coordinate region in competence-space
type Skill = axis in competence-space
type Habit = repeated function over time
type PainSignal = system signal
type Mistake = detected incompetence marker
type Correction = competence-restoring update
type Contradiction = corruption signal
type Construction = error-free build sequence
type Competence = quantified functional capacity
type Incompetence = quantified missing constructive capacity
type BasisSpan = transferable competence basis across contexts
```
```

4. Boolean arithmetic

```
```mhcd
A + B // OR, union, either condition accepted
```



```
A * B // AND, both conditions enforced
!A // NOT
A -> B // implication
A == B // equivalence
A != B // non-equivalence
```
```

Example:

```
```mhcd
rule R001:
 AXIOM[004] * AXIOM[005] -> operation.learn.valid
```
```

Meaning:

If `I can learn` and `Learning increases competence`, then `learn` is a valid update operation.

5. Lambda calculus layer

MHCDScript v2 supports lambda-style reusable operations.

```
```mhcd
lambda learn(x):
 ingest x
 encode x
 index x
 construct x
 validate x
 update competence
 return competence_delta(x)
```
```

```
```mhcd
lambda correct_without_guilt(error):
 enforce AXIOM[014]
 mark error as update_requirement
 locate missing_basis(error)
 construct correction(error)
 validate correction(error)
 return competence_restored
```
```

```
```mhcd
lambda optimize_context(skill, context):
 measure competence(skill, context)
 measure incompetence(skill, context)
 construct basis_improvement(skill, context)
 validate basis_improvement
 update competence_space
 return optimized(skill, context)
```
```

6. Program structure

Every valid v2 program has this form:

```
```mhcd
program "Program Name" {
 use FULL_MHCD

 define subject
 define contexts
 define skills
 define tools
 define records

 execute operations

 verify constructive_integrity
 enforce {014 + 015}
}
```
```

Minimal valid program:

```
```mhcd
program "Minimum MHCD Runtime" {
 use FULL_MHCD

 subject self = "Dominic Alexander Cooper"

 ingest current_state
 detect pain
 measure competence
 measure incompetence
 construct correction
 validate correction
 update competence_space

 enforce {014 + 015}
}
```
```

7. State model

The MHCD runtime is a state machine.

```
```mhcd
state MHCD_STATE {
 subject: Subject
 mind: Mind
 body: Body
 environment: Environment
 tools: Set<Tool>

 records: Set<Record>
 contexts: Set<Context>
}
```



```

 skills: Set<Skill>
 habits: Set<Habit>
 schedules: Set<Calendar>

 competence: Map<Skill, Number>
 incompetence: Map<Skill, Number>

 pain_signals: List<PainSignal>
 mistakes: List<Mistake>
 contradictions: List<Contradiction>
 corrections: List<Correction>

 enforced_axioms: Set<AxiomID>
}
...

```

A state transition has the form:

```

```mhcd
transition NAME(input_state) -> output_state {
    precondition condition
    operation operations
    postcondition condition
}
...

```

Example:

```

```mhcd
transition LEARN(input_state) -> output_state {
 precondition AXIOM[004] * AXIOM[005]

 operation {
 ingest input
 encode input
 index input
 construct usable_record
 validate usable_record
 update competence
 }

 postcondition competence(output_state) >= competence(input_state)
}
...

```

## 8. Core operations

### `ingest`

```

```mhcd
ingest source
ingest source as type
ingest source into context
...

```

Examples:

```
```mhcd
ingest "textbook chapter" as Note into "mathematics"
ingest "pain after failed proof" as PainSignal into "proof writing"
ingest "work feedback" as Record into "professional competence"
```
```

`encode`

```
```mhcd
encode input as structure
```
```

Examples:

```
```mhcd
encode "ring" as definition + example + non_example
encode "daily exercise" as habit
encode "project architecture" as diagram
```
```

`index`

```
```mhcd
index record with tags
index record under context
index skill on axis
```
```

Examples:

```
```mhcd
index "ideal" with ["ring theory", "substructure", "quotient"]
index "proof writing" under context "mathematics"
index "attention control" on axis "cognitive regulation"
```
```

`query`

```
```mhcd
query question
query question from records
query question under context
```
```

Example:

```
```mhcd
query "What is the weakest skill axis today?" under context "mathematics"
```
```

`construct`

```
```mhcd
construct object
construct object from components
construct object by steps
```
```

Example:

```
```mhcd
construct proof "kernel of ring homomorphism is ideal" by steps
```
```

`validate`

```
```mhcd
validate object
validate object against axioms
validate object by construction
```
```

Example:

```
```mhcd
validate proof against {008, 014, 015}
```
```

`detect`

```
```mhcd
detect pain
detect mistake
detect contradiction
detect incompetence
detect missing_basis
detect context_shift
```
```

Example:

```
```mhcd
detect incompetence in "proof construction"
```
```

`correct`

```
```mhcd
correct mistake
correct mistake with construction
correct incompetence by practice
```
```

Example:

```
```mhcd
correct "failed proof step" with direct construction
```
```

`measure`

```
```mhcd
measure competence
measure incompetence
measure competence in skill
measure competence in skill under context
```
```

Example:

```
```mhcd
measure competence in "applied ring theory" under context "quiet study"
```
```

`optimize`

```
```mhcd
optimize competence_basis
optimize skill across contexts
optimize basis_span for all_contexts
```
```

Example:

```
```mhcd
optimize "proof writing" across ["quiet study", "exam", "teaching", "coding"]
```
```

`externalize`

```
```mhcd
externalize record to tool
externalize memory to note
externalize intention to calendar
externalize procedure to checklist
```
```

Example:

```
```mhcd
externalize "weekly learning plan" to calendar
```
```

`schedule`

```
```mhcd
schedule action at time
schedule review every interval
schedule habit under context
```
```

...

Example:

```
```mhc  
schedule review "ring theory definitions" every 2 days
```
```

`execute`

```
```mhc  
execute plan
execute checklist
execute habit
execute construction
```
```

Example:

```
```mhc  
execute "daily competence plan"
```
```

`feedback`

```
```mhc  
feedback result
feedback result into record
feedback result updates competence_space
```
```

Example:

```
```mhc  
feedback "proof completed with two mistakes"
```
```

`enforce`

```
```mhc  
enforce axiom
enforce axiom_set
enforce {014 + 015}
```
```

Example:

```
```mhc  
enforce {
 AXIOM[014]
 AXIOM[015]
}
```
```

9. Records

A `record` is the universal database unit.

```
```mhcd
record ID {
 type: Note | Skill | Tool | Query | Answer | Habit | Correction | PainSignal | Context
 content: Text
 tags: List<Text>
 context: Context
 source: Text
 validated: Bool
 competence_effect: Number
 incompetence_effect: Number
}
```
```

Example:

```
```mhcd
record "MATH-RING-001" {
 type: Note
 content: "A ring is an algebraic structure with addition and multiplication."
 tags: ["algebra", "ring theory", "definition"]
 context: "mathematics"
 source: "constructed study note"
 validated: 1
 competence_effect: +0.05
 incompetence_effect: -0.02
}
```
```

10. Competence-space

Competence-space is an n-dimensional functional axial space.

```
```mhcd
competence_space "SELF" {
 axis memory
 axis attention
 axis reasoning
 axis proof_construction
 axis language
 axis tool_use
 axis bodily_energy
 axis emotional_regulation
 axis environment_design
 axis context_switching
}
```
```

Skills are embedded in competence-space:

```
```mhcd
```



```

skill "applied ring theory" {
 axes: [memory, reasoning, proof_construction, language]
 context: "mathematics"
 current_competence: 0.35
 target_competence: 0.90
}
...

```

Competence can be context-indexed:

```

```mhcd
context "quiet room" {
  modifies "applied ring theory" by +0.15
}

context "sleep deprivation" {
  modifies "applied ring theory" by -0.30
}
...

```

11. Error and contradiction handling

A mistake is never treated as guilt.

```

```mhcd
handler on_mistake(m):
 enforce AXIOM[014]
 mark m as detected_incompetence
 query "What basis is missing?"
 construct correction
 validate correction
 update competence_space
...

```

Contradiction is treated as a corruption signal.

```

```mhcd
handler on_contradiction(c):
  enforce AXIOM[006]
  mark c as corruption_signal
  quarantine c
  construct direct_alternative
  validate direct_alternative
...

```

Pain is treated as a system signal.

```

```mhcd
handler on_pain(p):
 enforce AXIOM[018]
 classify p

 if p.cause == incompetence:
 correct p.cause

```

```

 if p.cause == overload:
 reduce load

 if p.cause == contradiction:
 quarantine contradiction

 if p.cause == environment:
 modify context
...

12. Verification system

A script must pass four checks.

```mhc
verify script {
    check axiom_consistency
    check constructive_preference
    check innocence_preservation
    check competence_update_path
}
...

### Axiom consistency

```mhc
check axiom_consistency:
 no operation may deny AXIOM[014]
 no operation may deny AXIOM[015]
 no operation may treat contradiction as preferred learning
...

Constructive preference

```mhc
check constructive_preference:
    if proof_method == contradiction:
        reject as learning_engine
    if proof_method == construction:
        accept as learning_engine
...

### Innocence preservation

```mhc
check innocence_preservation:
 if mistake detected:
 require correction_without_guilt
...

Competence update path

```mhc

```



```
check_competence_update_path:
    if incompetence_detected:
        require_measurable_correction
...
```

13. General-purpose control flow

```
```mhc
if condition:
 operation
else:
 operation
...
```

```
```mhc
while condition:
    operation
...
```

```
```mhc
for item in collection:
 operation item
...
```

```
```mhc
try_construct object:
    construct object
    validate object
on_error error:
    correct error
...
```

Example:

```
```mhc
for skill in all_skills:
 measure_competence_in_skill
 if incompetence(skill) > 0.30:
 correct_skill_by_deliberate_practice
...
```

## 14. Modules

```
```mhc
module "Learning" {
    export learn
    export practice
    export review
    export validate_understanding
}
...
```

```
```mhc
module "Health" {
```

```

 export detect_energy
 export schedule_exercise
 export reduce_overload
}
...

```

```

```mhcd
module "Mathematics" {
    export define
    export prove_by_construction
    export generate_examples
    export validate_proof
}
...

```

15. Example program: daily runtime

```

```mhcd
program "MHCD Daily Runtime v2" {
 use FULL_MHCD
 use module "Learning"
 use module "Health"

 subject self = "Dominic Alexander Cooper"

 context current = detect context
 pain current_pain = detect pain
 energy current_energy = detect energy

 measure competence in all_skills under current
 measure incompetence in all_skills under current

 query "What competence update gives the greatest pain reduction today?"

 construct plan "daily competence plan" {
 item health_action
 item learning_action
 item work_action
 item tool_action
 item reflection_action
 }

 validate plan by construction
 externalize plan to checklist
 schedule review "daily competence plan" at "evening"

 execute checklist

 feedback result into competence_space

 enforce {014 + 015}
}
...

```

## 16. Example program: learning applied ring theory

```
```mhcd
program "Learn Applied Ring Theory v2" {
  use FULL_MHCD
  use module "Mathematics"

  skill "applied ring theory" {
    axes: [memory, reasoning, proof_construction, language, tool_use]
    context: "mathematics"
    current_competence: measure
    target_competence: 0.90
  }

  ingest "definition of ring"
  encode "ring" as definition + example + non_example
  index "ring" with ["algebra", "ring theory", "binary operations"]

  ingest "definition of ideal"
  encode "ideal" as definition + example + non_example
  index "ideal" with ["ring theory", "substructure", "quotient"]

  construct proof "kernel of ring homomorphism is ideal" by direct_steps
  validate proof by error_free_construction

  if detect contradiction:
    quarantine contradiction
    construct direct_alternative

  measure competence in "applied ring theory"
  correct detected_incompetence by deliberate_practice

  externalize summary to note
  schedule review "ring theory definitions" every "2 days"

  enforce {014 + 015}
}
```
```

## 17. Example program: contradiction quarantine

```
```mhcd
program "Contradiction Quarantine v2" {
  use FULL_MHCD

  input claim

  detect contradiction in claim

  if contradiction == true:
    mark claim as corruption_signal
    quarantine claim

    query "What direct constructive alternative exists?"
}
```

```

        construct alternative_proof
        validate alternative_proof by error_free_construction

        update records with alternative_proof
        update competence_space

    enforce {014 + 015}
}
...

```

18. Example program: competence optimiser

```

```mhcd
program "Competence Basis Optimizer v2" {
 use FULL_MHCD

 define contexts = [
 "quiet study",
 "public explanation",
 "exam condition",
 "work condition",
 "fatigue condition",
 "tool-supported condition"
]

 define skills = [
 "mathematics",
 "programming",
 "physical training",
 "attention control",
 "communication",
 "environment design"
]

 for skill in skills:
 for context in contexts:
 measure competence in skill under context
 measure incompetence in skill under context

 if incompetence(skill, context) > 0:
 construct basis_improvement for skill under context
 validate basis_improvement
 practice skill under context
 update competence_space

 optimize basis_span across contexts

 enforce {014 + 015}
}
...

```

## 19. Interpreter architecture

A real interpreter for MHCDScript v2 would need these components:

```
```mhcd
interpreter MHCDScriptV2 {
  lexer
  parser
  ast_builder
  axiom_registry
  type_checker
  state_engine
  verifier
  command_executor
  record_store
  competence_model
  context_model
  tool_bridge
  scheduler_bridge
  report_generator
}
```
```

Execution pipeline:

```
```mhcd
source_file
-> lexer
-> parser
-> abstract_syntax_tree
-> axiom_verifier
-> type_checker
-> state_transition_engine
-> constructive_validator
-> execution_log
-> updated_MHCD_state
```
```

## 20. File extensions

Recommended file extensions:

```
```text
.mhcd      main program file
.mhcda     axiom library
.mhcds     saved MHCD state
.mhcdlog   execution log
.mhcdmod   reusable module
.mhcdjson  JSON-compatible export
```
```

Example project structure:

```
```text
mhcd_project/
  main.mhcd
```
```

```

 axioms.mhcda
 state.mhcds
 modules/
 learning.mhcdmod
 mathematics.mhcdmod
 health.mhcdmod
 tools.mhcdmod
 logs/
 daily.mhcdlog
 exports/
 dashboard.mhcdjson
...

```

## 21. Machine-readable state export

```

```json
{
  "subject": "Dominic Alexander Cooper",
  "axioms": "000..063",
  "enforced": ["014", "015"],
  "competence_space": {
    "axes": [
      "memory",
      "attention",
      "reasoning",
      "proof_construction",
      "language",
      "tool_use",
      "bodily_energy",
      "emotional_regulation",
      "environment_design"
    ]
  },
  "runtime_rule": "learn_by_error_free_construction",
  "correction_rule": "mistake_is_update_requirement_not_guilt"
}
```

```

## 22. Core semantic law

The central semantic law of MHCDScript v2 is:

```

```mhc
semantic_law MHCD_CORE:
  existence
  * pain_signal
  * learning_capacity
  * measurable_competence
  * context_indexing
  * innocence
  * competence_space
  -> constructive_self_update_runtime
...

```


In compact Boolean form:

```
```mhcd
000 * 001 * 004 * 005 * 010 * 011 * 012 * 013 * 014 * 015
-> MHCD_RUNTIME
```
```

In lambda form:

```
```mhcd
λself.
 Exist(self)
 * Pain(self)
 * Learn(self)
 * Quantify(Competence(self))
 * Quantify(Incompetence(self))
 * ContextIndex(Competence(self))
 * Innocent(all)
 * FunctionalAxisSpace(Competence(self))
-> MHCD_RUNTIME(self)
```
```

23. Final identity of MHCDScript v2

```
```mhcd
MHCDScript v2 =
 axiom-governed
+ typed
+ stateful
+ constructive
+ innocence-preserving
+ context-indexed
+ competence-measuring
+ incompetence-correcting
+ pain-signal-aware
+ contradiction-quarantining
+ tool-extensible
+ environment-integrated
+ schedulable
+ human-centric
```
```

The shortest expression of the language is:

```
```mhcd
detect pain;
locate incompetence;
correct without guilt;
learn by construction;
measure competence;
optimize across contexts;
enforce {014 + 015};
```
```

Version 2 is therefore not only a notation system. It is a programmable self-correction language for the mental human-centric database.